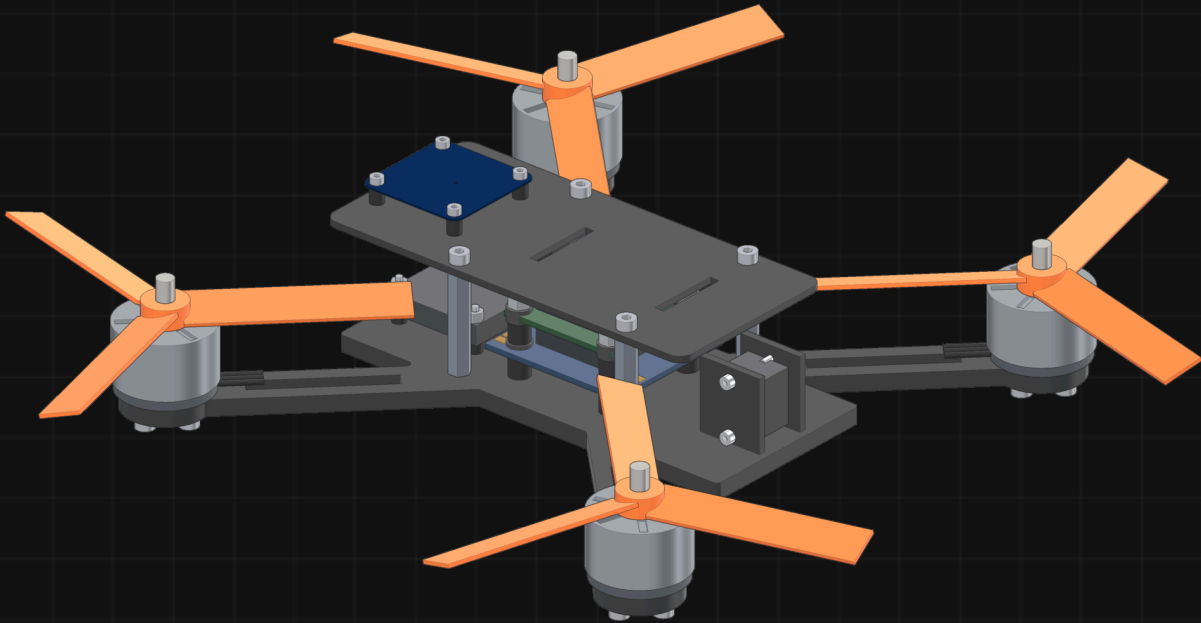


5-INCH 6S QUADCOPTER

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.



Assembled from UHD mates (16 placed bodies, 72 fasteners); strapped battery not shown.

STATIC THRUST 6.8 kg^D 4 motors, full throttle, maker's test prop	BATTERY 22.2 V 6S LiPo	CAPACITY 1100 mAh XT60
PEAK CURRENT 171.9 A 4 × motor 60 s peak	STATED MASS 425 g^D parts with a weight in the model	VIDEO LATENCY 20 ms DJI O4, Goggles 3

ABOUT THIS DOCUMENT

Every value is queried from the UHD model at build time and re-checked by `uhd-tech-docs-verify`. Markers show where a value is not a cited source value.

- ^D Derived: computed from model values; formula in the provenance appendix
 - ^S Stand-in: stand-in data derived from the model, not measured
 - Not in model: the model does not state this value (data gap)
- Unmarked values are cited from the part's sources (`sources.json`) or are design values of the model.

01 OVERVIEW

QUADCOPTER-SIN · V0.1.0

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.

ROLE	MODULE	QTY
Frame	5-inch quadcopter frame (custom)	1
Top plate	5-inch frame top plate (custom)	1
FC + ESC stack	DolphinRC F405 V3 + AM32 60A stack	1
Arms	Arm (motor + propeller)	4
6S battery	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	1
ELRS receiver	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	1
DJI O4 Air Unit	DJI O4 Air Unit	1
GNSS + compass	Matek M9N-5883 GNSS & Compass	1
Battery strap	HGLRC Kevlar battery strap 20x250mm	1
Bulk capacitor 470 µF	Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	1

CONTENTS

01	Overview	02
02	Specifications	03
03	Mechanical	04
04	Interfaces and pinout	06
05	Electrical characteristics	07
06	Performance	08
07	Design checks	10
08	Bill of materials	11
09	Evidence and assumptions	13
10	Value provenance	15

SYSTEM ARCHITECTURE

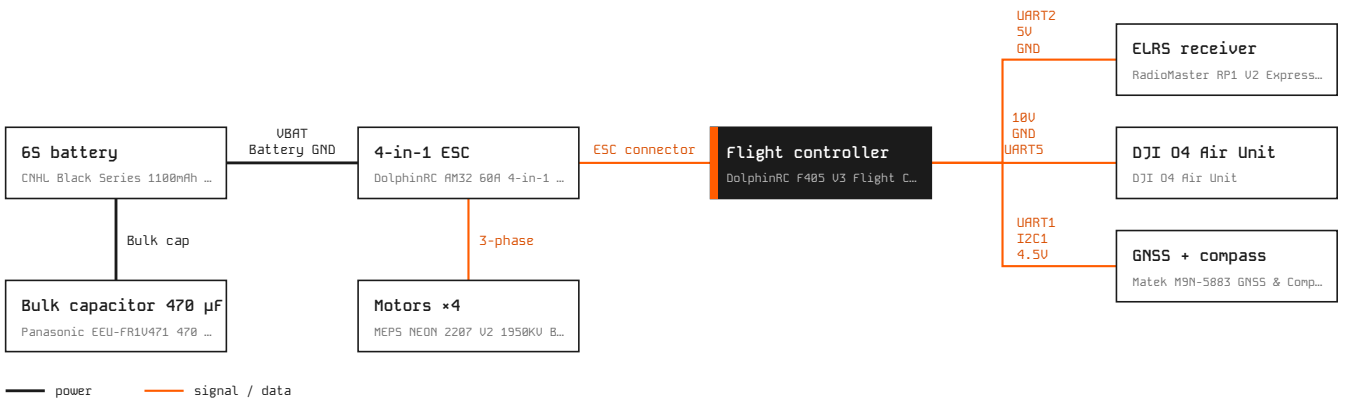


Figure 1 Power and signal links as stored in the model (validated with validateLinks); labels are the hub-side interfaces.

02 SPECIFICATIONS

PARAMETER	VALUE
AIRFRAME	
Layout	true-X, 5-inch
Wheelbase	225 mm
Bottom plate	5 mm
Standoff height	30 mm
Stack pattern square, M3	30.5 mm
Motor pattern bolt circle, M3	16 mm
PROPULSION	
Motor	MEPS NEON 2207 V2 1950KV Brushless Motor
KV	1950 rpm/V
Stator / poles	12N14P
Peak current (60 s)	42.97 A
Max power	1036 W
Motor mass	36 g
Propeller	HQProp Ethix S5 5x4x3 Propeller
Prop mass	3.7 g
ESC per channel continuous	60 A
ESC burst	65 A
ESC firmware	AM32 (32-bit); firmware target AT32DEVF421
POWER	
Battery	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)
Nominal voltage	22.2 V
Capacity	1100 mAh
Energy	24.42 Wh ^D
Continuous current	110 A
Burst current	220 A
Pack mass	211 g
ESC input	6–30 V
BEC 5 V rating	3 A
BEC 10 V rating	2.5 A
GPS rail	4.5–5.0 V

PARAMETER	VALUE
FLIGHT CONTROL	
Flight controller	DolphinRC F405 V3 Flight Controller
MCU	STM32F405RGT6, Cortex-M4 @ 168MHz
IMU	ICM-42688-P
Firmware target	Betaflight MATEKF405TE
UARTs	6
Receiver	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver
RC link	ExpressLRS
CRSF baud rate	420 000 Bd
VIDEO	
Air unit	DJI O4 Air Unit
Image sensor	1/2-inch CMOS
Field of view	117.6°
Max bitrate	100 Mbit/s
Supply range	3.7–13.2 V
Operating temperature	-10–40 °C
NAVIGATION	
GNSS module	Matek M9N-5883 GNSS & Compass
Receiver	u-blox NEO-M9N (concurrent reception of GPS, Galileo, GLONASS, BeiDou)
Compass address	0x2C
Supply current	50 mA

^D Derived: computed from model values; formula in the provenance appendix

^S Stand-in: stand-in data derived from the model, not measured – Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

03 MECHANICAL

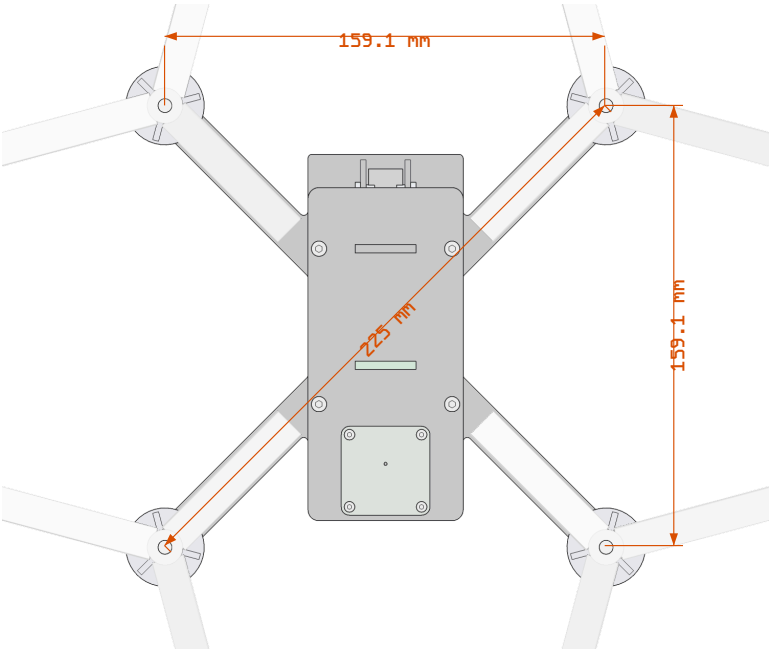


Figure 2 Top view, front up. Wheelbase is the frame's stated value; arm spacings are measured between the placed motor-mount frames. (unit: mm)

ENVELOPE

PARAMETER	VALUE
Wheelbase	225 mm
Arm spacing (x)	159.1 mm ^D
Arm spacing (y)	159.1 mm ^D
Plate thickness	5 mm
Standoff height	30 mm
Propeller diameter	5 in
Motor diameter	28.3 mm
Frame mass	--
Top plate mass	--

Frame material and mass are open data gaps of the custom frame module.

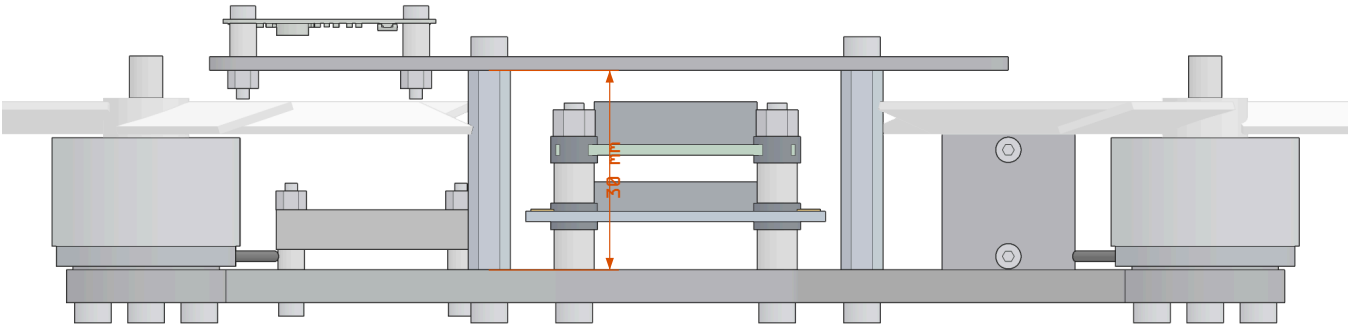


Figure 3 Right side, front to the right. Plate gap is the frame's standoff height. (unit: mm)

INTERFACE GEOMETRY

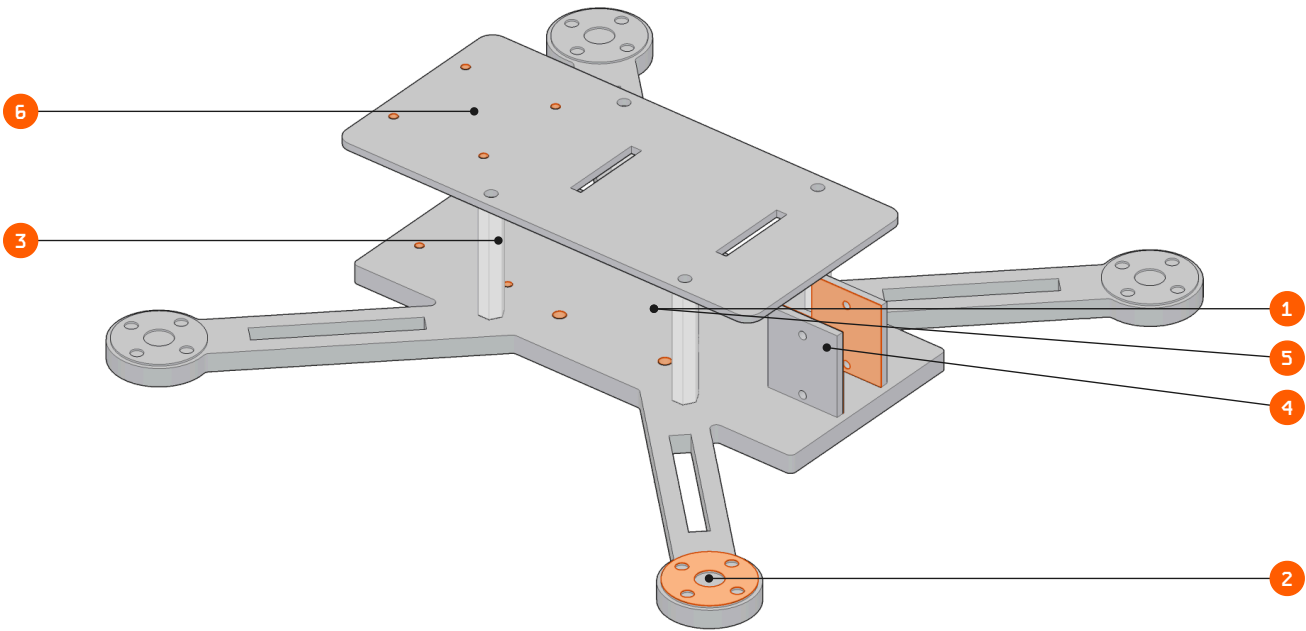


Figure 4 Structural interfaces of the frame and top plate (accent), from each interface's geometry frame and CAD feature binding.

#	INTERFACE	PATTERN	FASTENER	MATES WITH	BINDING
1	Stack mount frame.stack_mount	4 × on 30.5 mm square	3 mm	DolphinRC AM32 60A 4-in-1 ESC via stack_hardware DolphinRC F405 V3 Flight Controller via stack_hardware	D1 D2 D3
2	Front-right motor mount frame.motor_mount_fr	4 × on Ø16 mm circle	3 mm	MEPS NEON 2207 V2 1950KV Brushless Motor via arm_fr_hardware	D1 D2 D3
3	Transmission module mount frame.vtx_mount	4 × on 25.5 mm square	2 mm	DJI O4 Air Unit via vtx_hardware	D1 D2 D3
4	Camera side plates frame.camera_mount	4 × on 16 mm × 14 mm rectangle	2 mm	DJI O4 Air Unit via camera_hardware	D1 D2
5	Top-plate standoffs frame.standoff_mount	4 × on 56 mm × 48 mm rectangle	3 mm	5-inch frame top plate (custom) via frame_hardware	D1 D2 D3
6	GPS mount top_plate.gps_mount	4 × on 26 mm square	2 mm	Matek M9N-5883 GNSS & Compass via gps_hardware	D1 D2 D3

Binding: D1 named CAD feature, D2 per-interface artifact, D3 procedural from parameters (docs/geometry-artifacts.md).

04 INTERFACES AND PINOUT

Allocation of the DolphinRC F405 V3 Flight Controller (stack/fc) interfaces, pad to pad, from the stored links. Crossovers (TX→RX) are marked ✖. The link state is the result of pairwise DRC.

HUB INTERFACE	PADS (HUB → PEER)	PEER	PROTOCOL	PARAMETERS	STATE
UART2	RX2 → TX ✖ TX2 → RX ✖	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver CRSF (UART)	UART	baud rate 420 000 Bd	CONFIGURED
5V BEC	5V → 5V	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver 5V	POWER	voltage 5 V max current 3 A	CONFIGURED
GND	GND → G	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver G	POWER	—	CONFIGURED
10V/12V BEC	10V → VCC	DJI 04 Air Unit VCC (3-in-1 cable, red)	POWER	voltage 3.7–13.2 V max current 2.5 A	CONFIGURED
GND	GND → GND	DJI 04 Air Unit GND — power (3-in-1 cable, black)	POWER	—	CONFIGURED
UART5	R5 → TX ✖ TX5 → RX ✖	DJI 04 Air Unit UART (MSP DisplayPort / Canvas OSD)	UART	baud rate 115 200 Bd voltage 3.3 V	CONFIGURED
UART1	RX1 → TX ✖ TX1 → RX ✖	Matek M9N-5883 GNSS & Compass GNSS UART (NEO-M9N)	UART	baud rate 38 400 Bd	CONFIGURED
I2C1	SDA → DA SCL → CL	Matek M9N-5883 GNSS & Compass Compass I2C (QMC5883P / QMC5883L)	I2C	clock freq 100–400 kHz i2c address 0x2C	CONFIGURED
4.5V	4.5V → 5V	Matek M9N-5883 GNSS & Compass 5V	POWER	voltage 5 V	CONFIGURED
GND	GND → G	Matek M9N-5883 GNSS & Compass G	POWER	—	CONFIGURED

05 ELECTRICAL CHARACTERISTICS

BEC and rail utilisation (known loads)



Figure 5 checkSystem supply_budget.

Propulsion peak current vs battery ratings

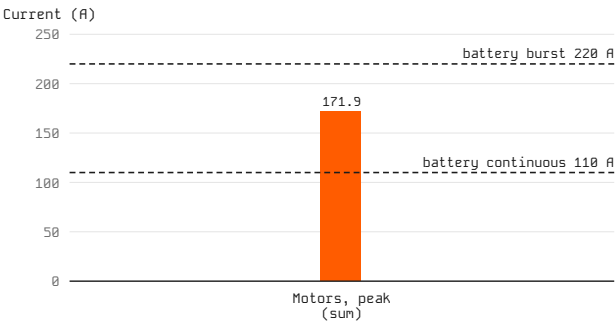


Figure 6 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.

POWER RAILS

RAIL	VOLTAGE	CAPACITY	KNOWN LOAD	LOADS	UNBUDGETED
5V BEC stack:bec_5v	5.0 V	15 W	0 W	receiver:vcc_5v	receiver:vcc_5v
10V/12V BEC stack:bec_10v	10.0 V	25 W	10 W	video:vcc	—
4.5V stack:rail_4v5	4.5–5.0 V	—	—	gnss:vin_5v	source has no current rating

From checkSystem rule supply_budget: capacity = rail voltage * current rating; loads that state no draw cannot be budgeted.

SUPPLY INPUTS

MODULE	INPUT	VOLTAGE RANGE	MIN. SUPPLY POWER
DolphinRC F405 V3 Flight Controller	VBAT VBAT	6.0–30.0 V	—
DolphinRC AM32 60A 4-in-1 ESC	BAT+ +	6.0–30.0 V	—
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	5V 5V	5.0 V	—
DJI O4 Air Unit	VCC (3-in-1 cable, red) VCC	3.7–13.2 V	10 W
Matek M9N-5883 GNSS & Compass	5V 5V	5.0 V	—
Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	+ lead +	0.0–35.0 V	—

Stored links: 28; configured: 28. Diagnostics on the nominal build: warning propulsion_current, info supply_budget, info supply_budget, info supply_budget.

06 PERFORMANCE

TEST DATA

MOTOR THRUST TEST (MANUFACTURER)

PROPELLER	SUPPLY	50 % THRUST	50 % CURRENT	100 % THRUST	100 % CURRENT	100 % POWER	RPM AT 100 %
MEPS S25145	25.2 V	763 g	10.4 A	1699 g	43.0 A	1036 W	30 342 rpm
MEPS S24942	25.2 V	648 g	9.5 A	1584 g	41.8 A	1009 W	31 973 rpm

MEPS NEON 2207 V2 1950KV Brushless Motor; the model holds the 50 % and 100 % rows of the maker's table. The fitted propeller (HQProp Ethix 55 5x4x3 Propeller) was not tested by the maker.

CAUTION Charts and figures tagged **STAND-IN • NOT MEASURED** are derived from the model to show the expected shape. Replace them by dropping measured files into `library/systems/quadcopter-5in/docs/test-data/` (status measured); the next build swaps them in.

Thrust and current per motor vs throttle

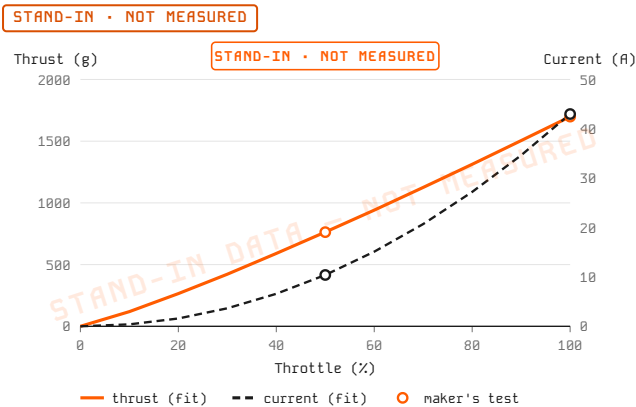


Figure 7 Circles: maker's test points (MEPS S25145). Lines: Stand-in: power-law fit through the motor maker's 50 % and 100 % test points, scaled to all motors, plus avionics draw.

Flight time by regime

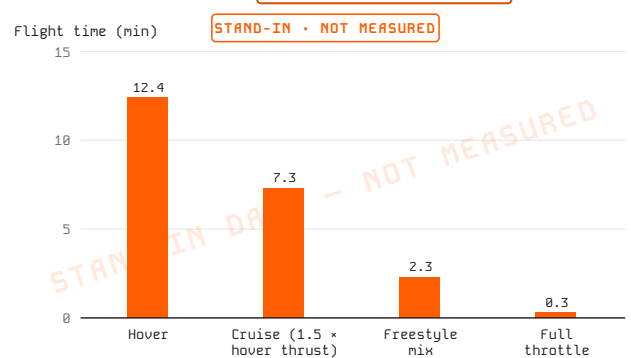


Figure 8 Stand-in: $\text{time} = \text{capacity} \times \text{usable fraction} \div \text{average current}$. Static-thrust currents: pessimistic for forward flight.

Battery discharge

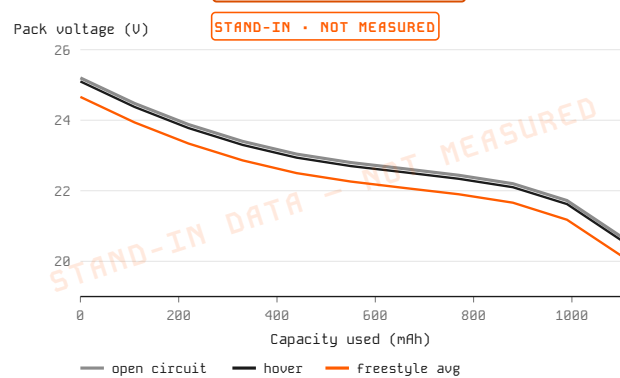


Figure 9 Stand-in: generic LiPo open-circuit curve per cell minus $I \cdot R$ sag, scaled to the pack.

ESC temperature over a punch-out

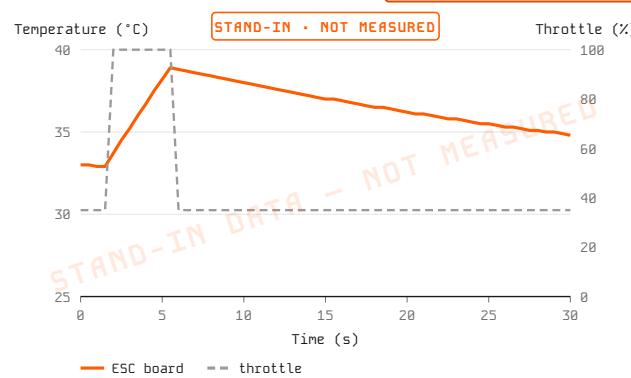


Figure 10 Stand-in: lumped thermal model $C \cdot dT/dt = P - (T - \text{ambient})/R$ with conduction and switching losses per channel.

PERFORMANCE SUMMARY

FIGURE	VALUE	TEST DATA
All-up weight (estimate)	550 g ^S	hover-current
Hover current	4.3 A ^S	hover-current
Hover power	95 W ^S	hover-current
Avionics current	0.7 A ^S	hover-current
Total static thrust at 100 %	6.80 kg ^S	throttle-sweep
Total current at 100 %	172.7 A ^S	throttle-sweep
Thrust-to-weight at 25.2 V ^S	12.4 ^S : 1	thrust-to-weight
Thrust-to-weight at 22.2 V ^S	9.6 ^S : 1	thrust-to-weight
Thrust-to-weight at 21.0 V ^S	8.6 ^S : 1	thrust-to-weight
Flight time, hover	12.4 min ^S	flight-time
Flight time, freestyle mix	2.3 min ^S	flight-time
ESC peak temperature	38.9 °C ^S	esc-temp-punchout

STAND-IN ASSUMPTIONS

- power laws through the maker's 50 % and 100 % points: thrust ~ throttle^{1.155}, current ~ throttle^{2.048}
- maker's test used the MEPS S25145 prop at 25.2 V, not the fitted propeller
- avionics: DJI O4 minimum supply power 10 W, GNSS 50 mA at 5 V, FC + receiver 2.5 W (assumed), BEC efficiency 85 % (assumed)
- all-up weight = 425.1 g of stated masses + 125 g assumed for the 67 parts without a mass (frame, top plate, hardware; see derived:mass.missing_count)
- usable capacity 80 % of rated
- thermal resistance 2 K/W and capacity 20 J/K (board in prop wash) – not in the model
- effective conduction resistance 4 m ohm per channel and 1.5 W switching loss per channel at full throttle – not in the model
- starting 8 °C above ambient after the previous flight segment
- generic LiPo open-circuit voltage curve
- internal resistance 4 m ohm per cell – not in the model

07 DESIGN CHECKS

CHECKSYSTEM

The nominal build and each fault scenario in `scenarios.ts` run through `checkSystem`. A fault scenario is expected to raise at least one error or warning.

SCENARIO	DIAGNOSTICS	FINDINGS (ERRORS AND WARNINGS)
Nominal build nominal	0 ERR 3 INFO	PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
04 wired to raw 6S o4-on-ubatt	2 ERR 2 INFO	LINK_STATE 04 10V: incompatible – voltage: 18–25.2 V vs 3.7–13.2 V (ranges do not overlap) HARNESS_CONNECTOR dji_cable.end_fc is dji_6pin but battery:battery_out is xt60. PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
20 × 20 frame for a 30.5 stack frame-20mm-stack	2 ERR 3 INFO	LINK_STATE ESC on frame: incompatible – hole_spacing: 30.5 mm vs 20 mm (ranges do not overlap); fastener_diameter: 3 mm vs 2 mm (ranges do not overlap) LINK_STATE FC on frame: incompatible – hole_spacing: 30.5 mm vs 20 mm (ranges do not overlap); fastener_diameter: 3 mm vs 2 mm (ranges do not overlap) PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
GPS on the receiver's UART gps-on-rx-uart	1 ERR 3 INFO	INTERFACE_REUSE stack:uart2 is linked 2 times (rx_crsf, gnss_uart) but supports 1. PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
Legacy GNSS (M10Q-5883, EOL) legacy-m10q	0 ERR 3 INFO	PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
Two compasses on one I2C bus two-compasses	1 ERR 3 INFO	BUS_ADDRESS stack:i2c1: 2 devices share I2C address 0x2C (gnss:i2c_compass, gnss_2:i2c_compass). PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.

08 BILL OF MATERIALS

GENERATED/BOM.JSON

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
1	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) cnhl-black-series-1100mah-6s-100c	CNHL (China Hobby Line) 1101006BK	1	PURCHASE	211 g	audited 18/18 A 3 · gaps 7
2	DJI 04 Air Unit dji-04-air-unit	DJI DF3L2904	1	PURCHASE	8.2 g	audited 32/34 A 1 · gaps 5
3	DolphinRC AM32 60A 4-in-1 ESC dolphinrc-am32-60a-4in1-esc	DolphinRC (Shenzhen DolphinRC Model Technology Co., Ltd.) DolphinRC AM32 ESC 60A (F405 V3 stack, SKU DPD200041 option '60A ESC')	1	PURCHASE	13.2 g	audited 18/18 A 1 · gaps 5
4	DolphinRC F405 V3 Flight Controller dolphinrc-f405-v3-flight-controller	DolphinRC (Shenzhen DolphinRC Model Technology Co., Ltd.) DolphinRC F405 V3 (F405 V3 stack, SKU DPD200041 option 'F405-FC')	1	PURCHASE	7.7 g	audited 28/30 A 1 · gaps 8
5	ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 ettinger-005-83-060-m3-nylon-spacer-6mm	ETTINGER 005.83.060	8	PURCHASE	0.15 g	audited 12/12 A 1 · gaps 1
6	HGLRC Kevlar battery strap 20x250mm hglrc-kevlar-battery-strap-20x250mm	HGLRC F.2D.02.00010	1	PURCHASE	8.3 g	audited 12/12 A 1 · gaps 5
7	HQProp Ethix S5 5x4x3 Propeller hqprop-ethix-s5	HQProp EthixS5-GR-PC	4 pack of 4	PURCHASE	3.7 g	audited 14/14 A 0 · gaps 4
8	M2 nyloc nut, DIN 985, A2 stainless iso-10511-m2-nyloc-nut	Westfield Fasteners WF16826	8	PURCHASE	—	— A 1 · gaps 0
9	M2 nylon spacer, Ø2.7/4.0 x 3 mm m2-nylon-spacer-3mm	Ettinger 007.02.403	4 pack of 100	PURCHASE	—	— A 1 · gaps 0
10	M2 nylon spacer, Ø2.7/4.0 x 5 mm m2-nylon-spacer-5mm	Ettinger 007.02.405	4 pack of 100	PURCHASE	—	— A 1 · gaps 0
11	M2 x 12 mm socket head cap screw, ISO 4762 iso-4762-m2x12	Generic (ISO 4762 / DIN 912) WF2332	4	PURCHASE	—	— A 1 · gaps 0
12	M2 x 18 mm socket head cap screw, ISO 4762 iso-4762-m2x18	Generic (ISO 4762 / DIN 912) WF17972	4	PURCHASE	—	— A 1 · gaps 0
13	M3 aluminium standoff, 30 mm, female-female, 5.5 mm Af m3-aluminium-standoff-30mm	Ettinger 005.03.308	4 pack of 100	PURCHASE	—	— A 1 · gaps 0
14	M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless iso-10511-m3-nyloc-nut	Generic (ISO 10511 / DIN 985) WF1288	4	PURCHASE	—	audited 12/12 A 1 · gaps 2
15	M3 x 30 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m3x30-socket-head-cap-screw	Generic (ISO 4762 / DIN 912) WF2352	4	PURCHASE	—	audited 12/12 A 1 · gaps 2
16	M3 x 8 mm socket head cap screw, ISO 4762 iso-4762-m3x8	Generic (ISO 4762 / DIN 912) WF2344	8	PURCHASE	—	— A 1 · gaps 0
17	Matek M9N-5883 GNSS & Compass matek-m9n-5883	Matek Systems M9N-5883	1	PURCHASE	14.5 g	audited 26/26 A 2 · gaps 8

Rows as generated by scripts/bom.ts (generated/bom.json). Mass is the part's stated weight; — = not in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
18	MEPS NEON 2207 V2 1950KV Brushless Motor meps-neon-2207-v2-1950kv	MEPS NEON 2207 V2 1950KV	4	PURCHASE	36 g	audited 20/20 A 0 · gaps 7
19	Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor panasonic-eeu-fr1v471	Panasonic Industry EEU-FR1V471	1	PURCHASE	—	audited 17/18 A 2 · gaps 3
20	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver radiomaster-rp1-v2-elrs-2g4	RadioMaster RP1 V2	1	PURCHASE	2.2 g	audited 21/21 A 0 · gaps 4
21	5-inch frame top plate (custom) quadcopter-5in-top-plate	Delta Robotics (custom) QUAD5-TOP	1	FABRICATE	—	— A 0 · gaps 1
22	5-inch quadcopter frame (custom) quadcopter-5in-frame	Delta Robotics (custom) QUAD5-FRAME	1	FABRICATE	—	— A 0 · gaps 1
23	DJI O4 3-in-1 cable dji-o4-3in1-cable	DJI	1	INCL. dji-o4-air-unit	—	— A 0 · gaps 0
24	FC/ESC SH1.0 8-pin cable dolphinrc-sh8-fc-esc-cable	DolphinRC	1	INCL. dolphinrc-f405-v3-flight-controller	—	— A 0 · gaps 0
25	M2 x 5 mm camera screw (included with the O4) o4-camera-screw-m2x5	Included (unbranded)	4	INCL. dji-o4-air-unit	—	— A 1 · gaps 1
26	M3 x 8 mm motor screw (included with the motor) motor-screw-m3x8	Included (unbranded)	16	INCL. meps-neon-2207-v2-1950kv	—	— A 1 · gaps 1
27	XT60 battery lead (12 AWG) dolphinrc-xt60-battery-lead	DolphinRC	1	INCL. dolphinrc-am32-60a-4in1-esc	—	— A 0 · gaps 0

Rows as generated by scripts/bom.ts (generated/bom.json). Mass is the part's stated weight; — = not in the model. A = assumption traits.

09 EVIDENCE AND ASSUMPTIONS

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	5	18/18	—	minimum (discharge cut-off) cell voltage; operating / charging / storage temperature; burst duration; main lead and balance lead length; XT60 gender (+2)
DJI O4 Air Unit	5	32/34	—	antenna connector family; FC-side 3-in-1 cable connector family; power consumption / max input current; storage temperature; lens mount in box
DolphinRC AM32 60A 4-in-1 ESC	3	18/18	—	pin 1 orientation of 8-pin SH connector (silkscreen order only); burst current duration (5 s vs 10 s) (sources conflict); operating temperature (not stated); VBAT passthrough current limit on connector V pin (not stated); current sensor output voltage range (only Betaflight scale/offset given)
DolphinRC F405 V3 Flight Controller	4	28/30	—	pin 1 orientation of all SH connectors (Manual shows silkscreen order only.); which UART the SBUS pad / DJI Sbus pin feeds, and inversion (Not stated; Betaflight target defaults serial RX to UART2.); 'ppm' silkscreen near RX3 pad (Low-res photo; target puts PPM on UART2 RX (PA3). Not claimed.); S1/S2 pad function (Product page says 10x PWM outputs (8 DShot), target defines SERVO1-3; S1/S2 modelled as PWM outputs only.); RSSI/Airspeed ADC pads (Listed as ADC channels but no pad labelled RSSI/Airspeed in drawings; omitted.) (+3)
ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25	3	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommeted board clamp height 4.0 mm each (ESC and FC)	operating temperature
HGLRC Kevlar battery strap 20x250mm	3	12/12	—	strap thickness; tensile strength / load rating; buckle dimensions and anti-slip strip length; temperature range; frame top-plate thickness and slot width
HQProp Ethix S5 5x4x3 Propeller	2	14/14	—	hub mounting style / retention; recommended motor / cell count; operating temperature; thrust / efficiency data
M2 nyloc nut, DIN 985, A2 stainless	—	not audited	A part choice: M2 nyloc nut, DIN 985, A2 stainless	—
M2 nylon spacer, Ø2.7/4.0 x 3 mm	—	not audited	A part choice: M2 nylon spacer, Ø2.7/4.0 x 3 mm	—
M2 nylon spacer, Ø2.7/4.0 x 5 mm	—	not audited	A part choice: M2 nylon spacer, Ø2.7/4.0 x 5 mm	—
M2 x 12 mm socket head cap screw, ISO 4762	—	not audited	A length: 12 mm	—
M2 x 18 mm socket head cap screw, ISO 4762	—	not audited	A length: 18 mm	—
M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF	—	not audited	A part choice: M3 aluminium standoff, 30 mm, female-female, 5.5 mm AF	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	2	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommets board clamp height 4.0 mm each (ESC and FC)	mass; nylon insert temperature limit
M3 x 30 mm socket head cap screw, ISO 4762, A2 stainless	2	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommets board clamp height 4.0 mm each (ESC and FC)	mass; operating temperature
M3 x 8 mm socket head cap screw, ISO 4762	—	not audited	A length: 8 mm	—
Matek M9N-5883 GNSS & Compass	13	26/26	—	pin 1 of JST-GH-6P; edge pads VBus, D-, D+, Bt and V-USB jumper; fastener size for 2 mm holes; hole pattern Y spacing; logic levels, I2C pull-ups on module (+3)
MEPS NEON 2207 V2 1950KV Brushless Motor	5	20/20	—	lead wire gauge (AWG); continuous current rating; operating / storage temperature; maximum voltage (V); prop nut thread handedness / shaft rotation (+2)
Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	3	17/18	A identity: Panasonic EEU-FR1V471 (Ø10 x 16 mm) A lead polarity identification: negative lead marked by the sleeve stripe	bundled capacitor brand/series/MPN; mass; ESR (as such)
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	8	21/21	—	supply current; operating temperature; logic level of TX/RX pads; receiver sensitivity (RP1-specific)
5-inch frame top plate (custom)	—	not audited	—	material; mass
5-inch quadcopter frame (custom)	—	not audited	—	material; mass
DJI O4 3-in-1 cable	—	not audited	—	—
FC/ESC SH1.0 8-pin cable	—	not audited	—	—
M2 x 5 mm camera screw (included with the O4)	—	not audited	A length: 5 mm	screw length
M3 x 8 mm motor screw (included with the motor)	—	not audited	A length: 8 mm	screw length
XT60 battery lead (12 AWG)	—	not audited	—	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

10 VALUE PROVENANCE

This document shows 404 bound values. Each is a query into the UHD model, re-resolved and compared by `uhd-tech-docs-verify`; the full list with sources is in `uhd-quad5-ds.values.json` next to the document.

STATUS	VALUES	MEANING
Cited source	86	stated by a source in the part's <code>sources.json</code>
Model design value	263	a design value of a custom module, or a UHD check result
Derived	24	computed from model values; formula in the provenance appendix
Stand-in	15	stand-in data derived from the model, not measured
Not in model	16	the model does not state this value (data gap)

STATUS	VALUE	QUERY	DERIVATION / REASON
D Derived	6.8 kg	<code>derived:propulsion.thrust_full_total</code>	<code>motors * performance.thrust_tests[0].full_throttle.thrust_g * manufacturer test with MEPS S25145 at 25.2 V * inputs: 1</code>
D Derived	425 g	<code>derived:mass.known_total</code>	$\Sigma (\text{count} \times \text{domains}[\text{mechanical}].\text{weight_g})$ over leaf instances and placed hardware <code>* inputs: 10</code>
D Derived	24.42 Wh	<code>derived:power.battery_energy_Wh</code>	<code>battery voltage * capacity * inputs: 2</code>
D Derived	159.1 mm	<code>derived:frameDistance(frame.motor_mount_fl, frame.motor_mount_fr)</code>	<code> placement(a) * a.frame.origin - placement(b) * b.frame.origin </code> from <code>assemble()</code> <code>* inputs: 2</code>
D Derived	159.1 mm	<code>derived:frameDistance(frame.motor_mount_rr, frame.motor_mount_fr)</code>	<code> placement(a) * a.frame.origin - placement(b) * b.frame.origin </code> from <code>assemble()</code> <code>* inputs: 2</code>
D Derived	159.1 mm	<code>derived:frameDistance(frame.motor_mount_fl, frame.motor_mount_rl)</code>	<code> placement(a) * a.frame.origin - placement(b) * b.frame.origin </code> from <code>assemble()</code> <code>* inputs: 2</code>
- Not in model	—	<code>def:quadcopter-5in-frame:domains[domain=mechanical].weight_g</code>	
- Not in model	—	<code>def:quadcopter-5in-top-plate:domains[domain=mechanical].weight_g</code>	
S Stand-in	550 g	<code>test:hover-current:summary[id=auw_g].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/hover-current.json</code>
S Stand-in	4.3 A	<code>test:hover-current:summary[id=hover_current_A].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/hover-current.json</code>
S Stand-in	95 W	<code>test:hover-current:summary[id=hover_power_W].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/hover-current.json</code>
S Stand-in	0.7 A	<code>test:hover-current:summary[id=avionics_current_A].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/hover-current.json</code>
S Stand-in	6.80 kg	<code>test:throttle-sweep:summary[id=thrust_total_full].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/throttle-sweep.json</code>
S Stand-in	172.7 A	<code>test:throttle-sweep:summary[id=current_total_full].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/throttle-sweep.json</code>
S Stand-in	25.2 V	<code>test:thrust-to-weight:rows[0][0]</code>	<code>library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json</code>
S Stand-in	12.4	<code>test:thrust-to-weight:rows[0][2]</code>	<code>library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json</code>
S Stand-in	22.2 V	<code>test:thrust-to-weight:rows[1][0]</code>	<code>library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json</code>
S Stand-in	9.6	<code>test:thrust-to-weight:rows[1][2]</code>	<code>library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json</code>
S Stand-in	21.0 V	<code>test:thrust-to-weight:rows[2][0]</code>	<code>library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json</code>
S Stand-in	8.6	<code>test:thrust-to-weight:rows[2][2]</code>	<code>library/systems/quadcopter-5in/docs/test-data/thrust-to-weight.json</code>
S Stand-in	12.4 min	<code>test:flight-time:summary[id=hover_min].value</code>	<code>library/systems/quadcopter-5in/docs/test-data/flight-time.json</code>

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
S Stand-in	2.3 min	test:flight-time:summary[id=mix_min].value	library/systems/quadcopter-5in/docs/test-data/flight-time.json
S Stand-in	38.9 °C	test:esc-temp-punchout:summary[id=peak_C].value	library/systems/quadcopter-5in/docs/test-data/esc-temp-punchout.json
D Derived	0	derived:diagCount(nominal,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(nominal,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	3	derived:diagCount(nominal,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	2	derived:diagCount(o4-on-vbat,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(o4-on-vbat,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	2	derived:diagCount(o4-on-vbat,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	2	derived:diagCount(frame-20mm-stack,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(frame-20mm-stack,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	3	derived:diagCount(frame-20mm-stack,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(gps-on-rx-uart,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(gps-on-rx-uart,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	3	derived:diagCount(gps-on-rx-uart,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	0	derived:diagCount(legacy-m10q,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(legacy-m10q,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	3	derived:diagCount(legacy-m10q,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(two-compasses,error)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	1	derived:diagCount(two-compasses,warning)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
D Derived	3	derived:diagCount(two-compasses,info)	count of checkSystem(scenario).diagnostics with the severity · inputs: 1
– Not in model	–	sys:bom[line=8].verification	UHD bom
– Not in model	–	sys:bom[line=9].verification	UHD bom
– Not in model	–	sys:bom[line=10].verification	UHD bom
– Not in model	–	sys:bom[line=11].verification	UHD bom
– Not in model	–	sys:bom[line=12].verification	UHD bom
– Not in model	–	sys:bom[line=13].verification	UHD bom
– Not in model	–	sys:bom[line=16].verification	UHD bom
– Not in model	–	sys:bom[line=21].verification	UHD bom

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
- Not in model	-	sys:bom[line=22].verification	UHD bom
- Not in model	-	sys:bom[line=23].verification	UHD bom
- Not in model	-	sys:bom[line=24].verification	UHD bom
- Not in model	-	sys:bom[line=25].verification	UHD bom
- Not in model	-	sys:bom[line=26].verification	UHD bom
- Not in model	-	sys:bom[line=27].verification	UHD bom